IMAGE STEGANOGRAPHY USING CNN

A PROJECT REPORT

for

INFORMATION SECURITY MANAGEMENT

in

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by

RITWEEK RAJ (19BCE0597)

PANKAJ SHARMA (19BCE0572)

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Prof. SIVA SHANMUGAM G

School of Computer Science and Engineering

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Study on Image Steganography using CNN

RITWEEK RAJ[1], PANKAJ SHARMA[2]

[1,2] **Department of Computer Science and Engineering, VIT University, Vellore, Tamil Nadu, India**

Abstract

The unfamiliarity and curiosity in the subject were the key reasons for choosing steganography from the list of available project ideas. Another argument is that it ensures data security. Even if a hacker (or intruder) has access to our image files, they will be unable to access our information; that is, even if the intruder manages to hack and gain access to our system, they will be unable to access our data. This feature adds a lot to the appeal. The project is about utilising images to learn about text stenography. For images, image steganography is used, and the relevant data is decrypted in order to get the message image. Image steganography is researched, and one of the methods is used to demonstrate it, because it may be done in a variety of ways. Steganography is the practise of concealing data, such as text, photos, or audio files, within another image or video file. In a word, steganography's fundamental goal is to conceal the desired information within any image, audio, or video that does not appear to be hidden just by looking at it. Image-based Steganography is based on a basic concept. Images are made up of digital data (pixels) that define what's inside the image, which is usually the colours of all the pixels. Since we know that every image is made up of pixels, each of which has three values (red, green, blue). We anticipate that the image we want to conceal will be totally absorbed by the cover image, making it invisible to intruders.

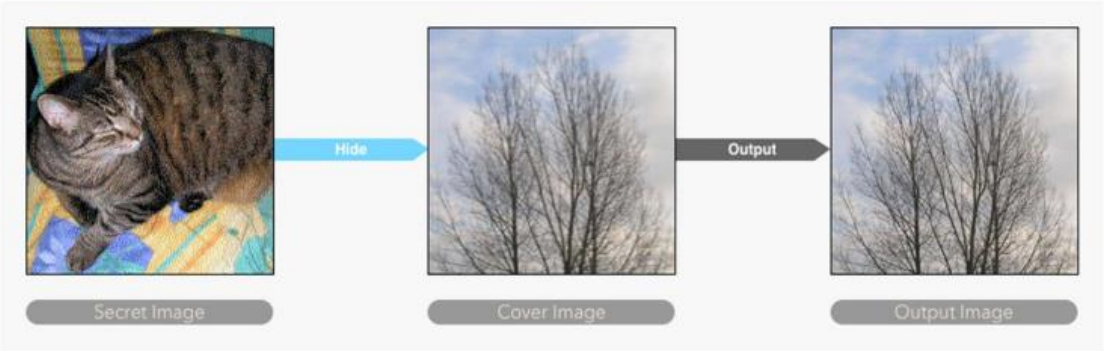
Keywords

Image Steganography, Steganography, Steganography using CNN, Deep learning

I. INTRODUCTION

A technique used for hiding secret data within a standard, public file or message to avoid detection is known as Steganography. The secret data is then extracted once it arrives at its destination. Steganography, in conjunction with encryption, can be used to further conceal or safeguard data.

Using deep convolutional neural networks, a full-sized colour image is concealed inside another image (called Cover image) with minimum changes in appearance. The hidden image will then be revealed by combining a "reveal" network with the created image.

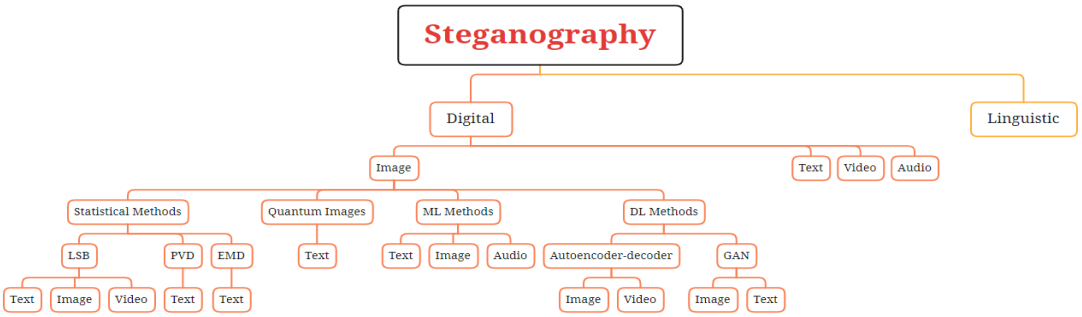


The first image in the above example is the secret image that must be kept concealed, while the second image is the cover image that will be used to conceal the secret image. The third image, which is essentially a repainted cover image, is the result of the hiding procedure. Some of the most common steganographic approaches involve the manipulation of the least significant bits (LSB) of the images to place the secret information, which can be achieved adaptively or even uniformly through simple replacement or through methods or through methods that are more advanced. In our project we will be using convolutional neural networks (CNN) for hiding an image inside another image. The advantage is a more efficient steganography.

There are disadvantages to this approach also, as on enhancing the image by a big factor, one can find out the original image but it is very hard to do this and to completely reconstruct the original image from the output image. We have used the TINY IMAGENET DATABASE.

Summary of various methods

Following a review of all available frameworks, the methodologies are primarily divided into three categories: traditional image steganography methods, CNN-based image steganography methods, and GAN-based image steganography methods. Traditional methods are frameworks that employ methods unrelated to machine learning or deep learning algorithms. The LSB technique is used in many traditional methods. CNN-based methods use deep convolutional neural networks to embed and extract secret messages, whereas GAN-based methods use some GAN variants.



Classification of current methods on basis of method and media use

II. BACKGROUND

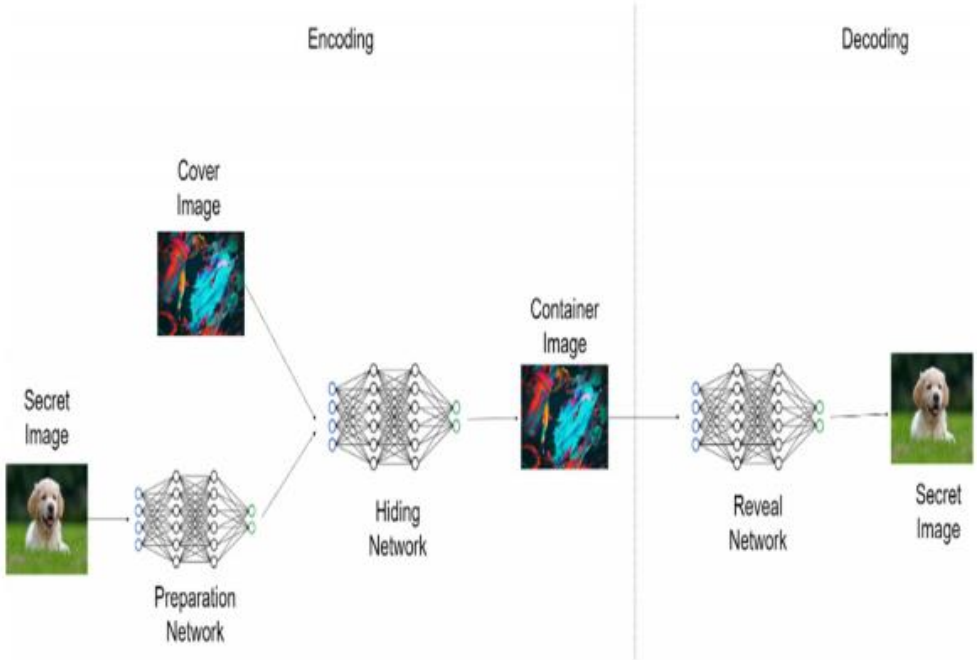
General Architecture on Topic chosen

The entire system consists of 3 CNNs:

1. The Preparation Network: This network prepares the secret image to be hidden the purpose is to transform the color-based pixels to more useful features so that in can be encoded such as edges. It contains 50 filters of (3X3, 4X4, 5X5) patches there are total 6 layers of this kind.

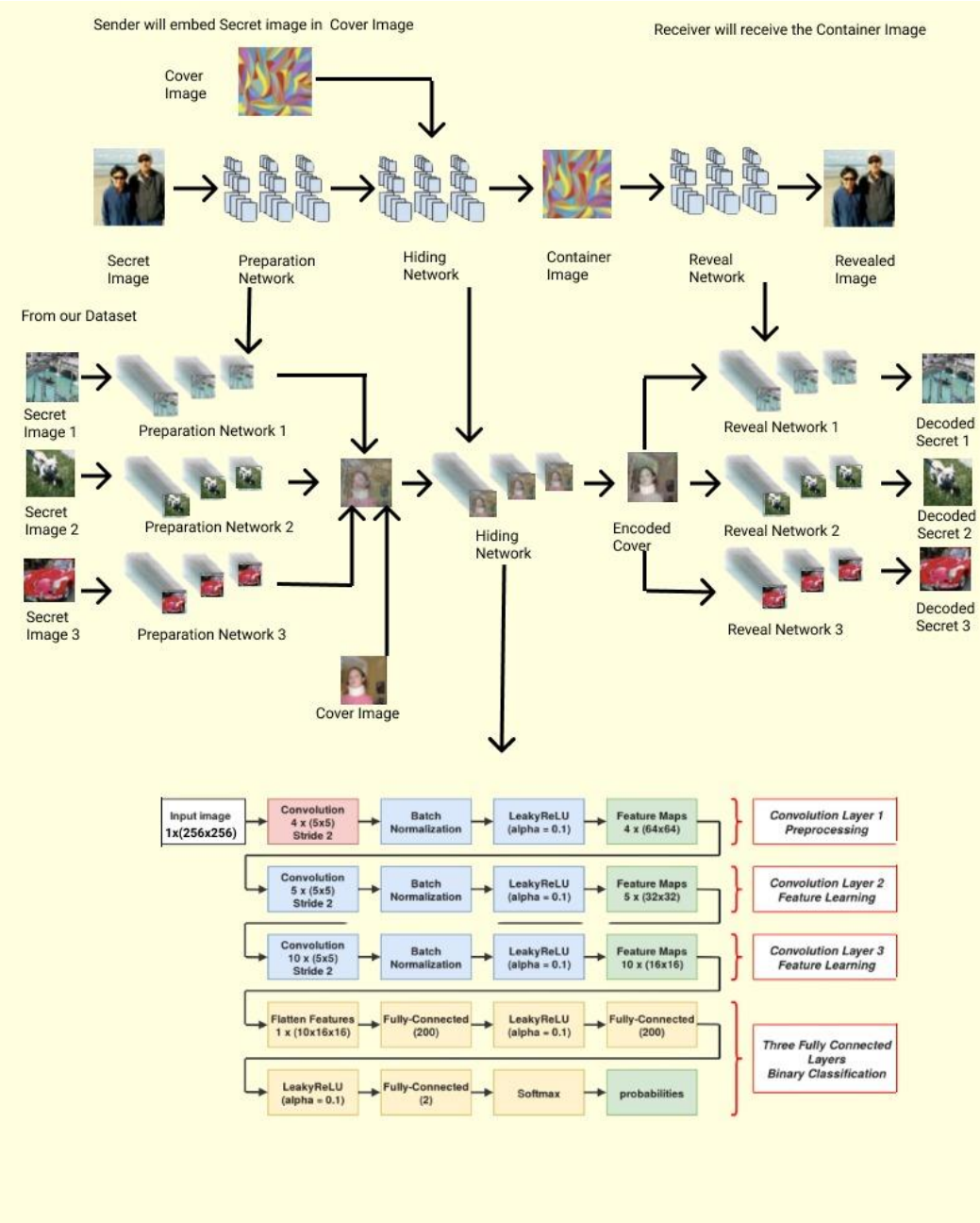
2. The Hiding Network: This network will take the output of the preparation network and will then create a Container Image. It is a CNN with 5 convolutional layers that have 50 filters of (3X3, 4X4, 5X5) patches. This layer consists of total of 15 convolutional layers in this network.

3. The Reveal Network: Converts the Container image to the original image this network is used for decoding. Which has 5 convolutional layers that have 50 filters of (3X3, 4X4, 5X5) patches. This layer consists of total of 15 convolutional layers in this network.



Some small amount of noise is also added to the images that are the output of the second network in order to make sure that the system does not simply encode the secret image in the LSB.

Architecture Diagram



We want to do image steganography, which involves hiding an image inside one cover image. The concealed images included in the cover image must be retrievable with minimal loss. The encoded cover image must resemble the original cover image in appearance.

We transfer secret images over the prep network, then concatenate the resulting data with the carrier image before sending it over the Hiding network. The idea of having decoders, one for each secret picture, to obtain the secret image from the container image is then implemented.

To make our image retrieval model more secure, we expand on the concept of putting instead, a secret image with noise in the original cover image placing the secret photos on the original cover's LSBs image.

Following is a brief overview of the encoder/decoder framework used in our technique:

ENCODER: A prep network that corresponds to secret image input. The outputs of the prep network are combined with the cover picture and passed via the Hiding network.

DECODER: The decoder network is made up of reveal networks, each of which has been taught to decode its own message.

The following is the typical framework:

Prep Networks: Each prep network is made up of two layers stacked on top of each other. Each layer is made up of three independent Conv2D layers. These three Conv2D layers have 50, 10, and 5 channels, respectively, with kernel sizes of 3, 4, and 5 for each layer. Along both axes, the stride length remains constant at one. To preserve the output image in the same dimensions, appropriate padding is provided to each Conv2D layer. After each Conv2d layer, a Relu activation is applied.

Concealing Network: The concealing network is a three-layer aggregation. Each of these layers is made up of three individual Conv2D layers. The Conv2D layers in the hidden network have a similar basic structure to the Conv2D levels in the Prep Network.

Reveal Network: The reveal network has a similar basic design to the hidden network, with three levels of Conv2D layers that are comparable in shape.

Novelty of our project

- ➔ Given any dataset of any size we will be normalizing the image on a scale of 255 X 255 so the size of the image will not affect the results, while in other models passing images of other sizes doesn't give the appropriate result

- ➔ Beta variable is declared to weight the losses of the secret and cover images and its value is set based on the most feasible value

- ➔ The function implemented by us returns the encoder as a Keras model, composed by Preparation and Hiding Networks which once hosted can used for encoding by anyone

- ➔ The function implemented by us returns the decoder as a Keras model, composed by the Reveal Network

- ➔ Here we are adding Gaussian noise with 0.01 standard deviation for better training accuracy and differential enhance magnitude of 1.0

- ➔ Our model is compatible with both grey scale and rgba color values, so the color of the image will not have an affect on the result and corresponding functions for the same have been defined in the code as well.

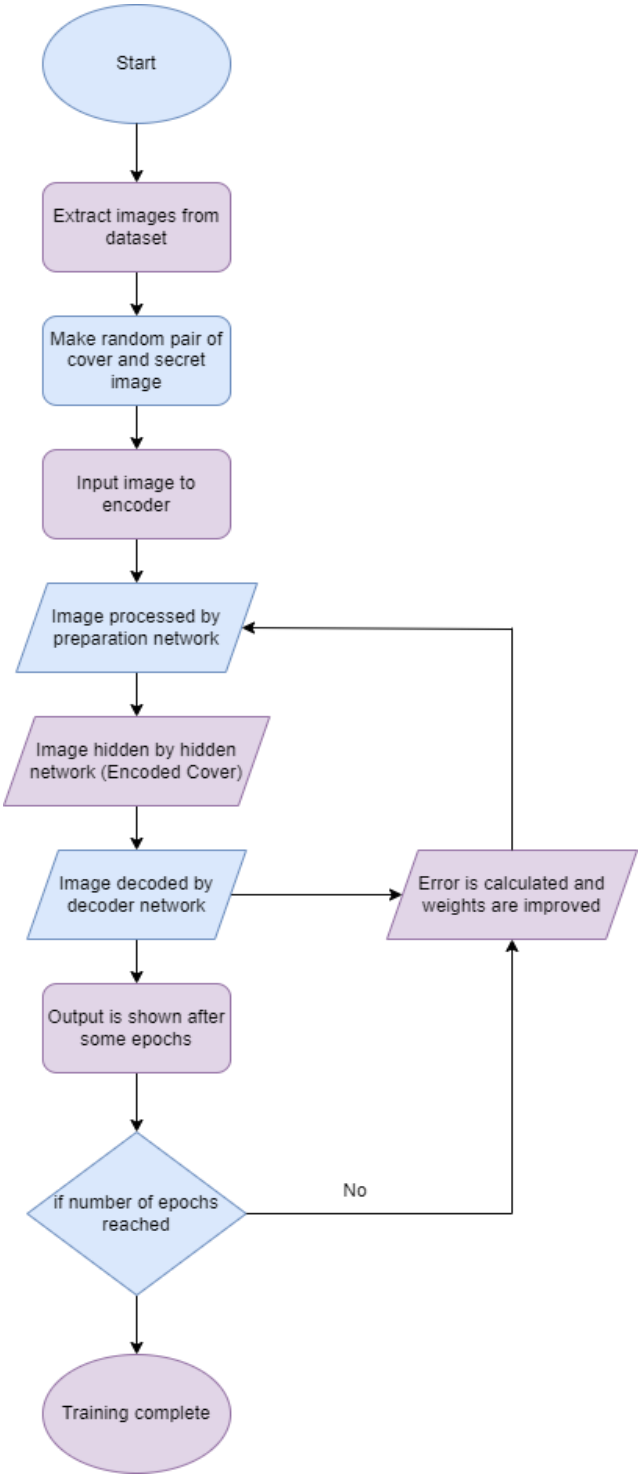
- ➔ Compared our model on three different activation functions RELU, SELU and TANH and added adam optimizer layer to improve training accuracy

- ➔ Used various matrix of comparison and have got accuracy better than majority of the models found by us during literature review. A comparison of the same is also shown in the table at the end of the document

CODE

https://github.com/ritweekraj2802/IMAGE-STEGANOGRAPHY-USING-CNN.git

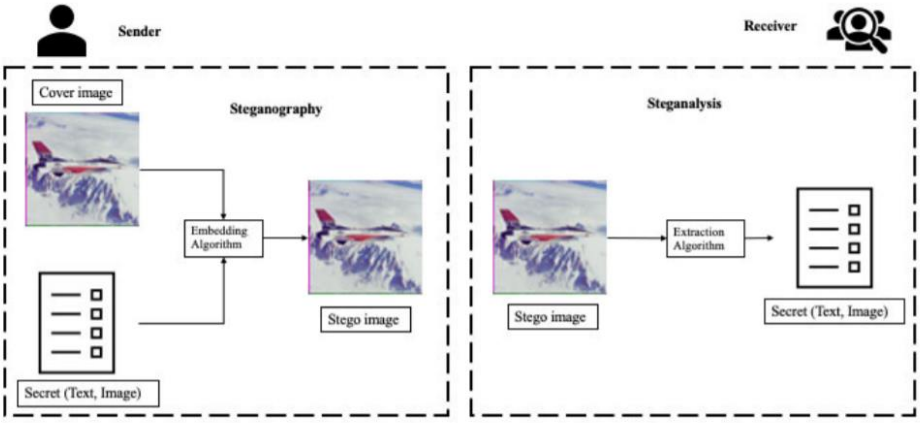
Flow Chart on Topic chosen



Review on various schemes

Image steganography is the process of concealing information in a cover image, which might be text, image, or video. The secret information is concealed in such a way that it cannot be seen by human sight. Deep learning technology, which has recently gained popularity as a strong tool in a variety of applications, including image steganography, has gotten a lot of attention. The primary goal of this study is to analyze and give an explanation for the numerous deep studying algorithms to be had inside the field of photograph steganography which we came across in our referred papers. Traditional methods, Convolutional Neural Network-based methods, and General Adversarial Network-based methods are the three types of deep learning approaches utilized for image steganography. This study includes a detailed explanation of the datasets used, experimental set-ups explored, and commonly used assessment measures, in addition to the methodology. For simple reference, a table summarizing all of the details is also supplied.

General principle of steganography



Steganography techniques that are based on tradition methods: -

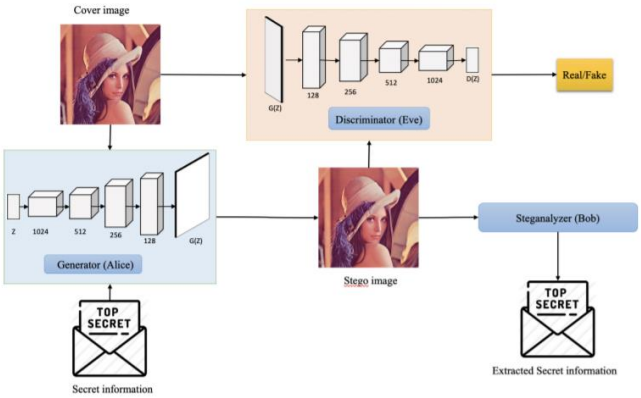
Image steganography is traditionally done using the Least Significant Bits (LSB) substitution method. The pixel quality of images is normally higher, but not all of the pixels are used. The LSB approach is based on the notion that changing a few pixel values will not result in noticeable changes. The secret data is transformed into a binary format. The least significant bits in the noisy area are determined by scanning the cover image. The LSBs of the cover picture are then replaced with the binary bits from the secret image. The substitute approach must be used with caution, as overloading the cover picture may result in noticeable alterations that reveal the secret information's presence.

Dataset	**Metrics**	**Advantages**	**Disadvantages**
Lena	PSNR	Time to compute is reduced. The image is a coded message in and of itself.	It is insecure.
Lena and Baboon	PSNR and MSE	Time to compute is reduced. The image is a coded message in and of itself. Any image format is acceptable.	When compared to deep learning approaches, security is a concern.
1 RGB Image	PSNR and Time	Time to compute is reduced. It embeds and extracts data with ease. Steganography and steganalysis are not reliant on one other.	It is insecure. Text is used to communicate secret information.

GAN – BASED methods used in Steganography: -

General Adversarial Networks (GAN) are a type of deep CNN. For image generation tasks, a GAN employs game theory to train a generative model with an adversarial process. In GAN architecture, two networks – generator and discriminator networks – compete to generate a perfect image. The data is fed into the generator model, and the output is a close approximation of the given input image. The discriminator networks classify the generated images as either false or true. The two networks are trained in such a way that the generator model attempts to imitate the input data as closely as possible with as little noise as possible.

Image steganography existing methods using a GAN architecture can be categorized into five types: a three-network-based GAN model, cycle-GAN-based architectures, sender receiver GAN architectures, coverless models in which the cover image is generated randomly rather than being given as input, and an Alice, Bob, and Eve based model.



A GAN model is made up of two main components: the generator and the discriminator. In some of the methods for image steganography, a new network called the steganalyzer is introduced. These three components' primary functions are as follows:

- *A model, G, for generating stego images from the cover image and the random message.*
- *A discriminator model, D, to determine whether the generated image from the generator is real or fake.*
- *A steganalyzer, S, to determine whether or not the input image contains confidential secret data.*

Architecture	Dataset	Advantages	Disadvantage
Alice, Bob, Eve	BOSSbase and celebA	Embedding does not necessitate domain knowledge.	Messages are used instead of images. Grid search for embedding scheme selection consumes time.
DCGAN	celebA and BOSSbase	Game-theoretic formulation is used	Steganalyzer is used to provide the probability by itself, which lead in additional computational cost and overhead.
GAN	CelebA	Unlimited number of cover images are be generated	Generator and discriminator are shared. Images that have been generated are not natural.
DCGAN	CelebA	Image creation that is more realistic Extremely safe	It provides probabilistic information rather than secret information. There is no information on how to obtain the secret data.

Steganography techniques that are based on CNN:-

The encoder-decoder architecture is greatly influenced by steganography employing CNN models. The encoder receives two inputs: the cover picture and the secret image, which are used to construct the stego image, and the decoder receives the stego image, which outputs the embedded secret image. The core principle remains the same, however different techniques have experimented with various structures. Different algorithms change in how the input cover picture and the secret image are concatenated, while variations in the convolutional layer and pooling layer are to be expected. The number of filters used, the strides taken, the filter size utilised, the activation function used, and the loss function used differ from one approach to the next. One thing to keep in mind is that the cover image and the secret image must be the same size, so that every pixel of the secret image is dispersed throughout the cover image.

Convolution is a type of linear operation that expresses the degree of overlap between two functions when they are shifted over each other. Convolutional networks are simple neural networks with at least one layer that uses convolution instead of ordinary matrix multiplication. The following are some of the benefits of using a CNN-based architecture for encoding and decoding:

- *CNN extracts visual features automatically.*
- *CNN essentially down samples the image using nearby pixel information, first by convolution, and then by using a prediction layer at the end.*
- *CNN is more accurate and performs effectively.*

One can get a decent idea of the patterns of natural images by using a deep neural network, in this case a CNN. The network will be able to determine which areas are redundant, allowing additional pixels to be buried in certain areas. The amount of hidden data can be raised by saving space on superfluous areas. Because the structure and weights can be randomized, the network will conceal data that is inaccessible to anyone who does not have the weights.

Architecture	**Dataset**	**Advantages**	**Disadvantages**
Encoder-decoder with SCR	ImageNet	Highly secure and dependable	The loss used isn't ideal. In black or white areas, visible noise might be seen.
Encoder-decoder with VGG-base	COCO and wikiart.org	It is not necessary to have prior domain knowledge. Since the created image is unrelated to the secret	High number of images are required in computation purpose.

		information, it is extremely secure.	
CNN	ImageNet and Holiday	The image is a coded message in and of itself. The simplest and most basic architecture is chosen. To speed up training, a new error back propagation function is implemented.	However, the image is only 64x64 pixels in size, which is quite little. The images in the input are simply concatenated.
U-Net	ImageNet	The image is a coded message in and of itself. The simplest and most basic architecture is chosen.	However, the image is only 64x64 pixels in size, which is quite little. The images in the input are simply concatenated.
Encoder-decoder	ImageNet	The image is a coded message in and of itself.	However, the image is only 64x64 pixels in size, which is quite little.

Comparison of dataset used in different papers

Dataset	**Image Format**	**Number of Samples**	**Image Size**	**Purpose**
BOSSBase	tiff	Training-9074 Testing-1000	512 x 512 x 1	Steganalysis and Steganography
CelebA	jpg	Greater than 200K	218 x 178 128 x 128	Landmark localization, Face editing synthesis, Face Detection

ImageNet	Arbitrary	Greater than 14M	Arbitrary	Computer Vision
MNIST Handwritten Digits	idx	Training-60000 Testing-10000	28 x 28 x 1	Computer Vision and Image Processing
COCO	jpg	330K	640 x 640 x 3	Image Captioning, segmentation and Object Detection
Div2K	png	Training-800 Testing-100 Validation-100	1020 x 678 x 3	Single Image Super Resolution
SZUBase	jpg	40000	512 x 512 x 1	Steganalysis and Steganography
USC-SIPI	tiff	170	256 x 256 x 1, 512 x 512 x 1, 1024 x 1024 x 1	Machine Vision, Image Analysis, and Processing
DTD	jpg	5640	300 x 300 x 3, 640 x 640 x 3	Textural Analysis
LFW	jpg	13233	150 x 150 x 3	Face Verification
Pascal VOC	jpg	11530	500 x 300 x 3	Object Classification and Detection

We will be using the Tiny ImageNet which contains 100000 images divided into 200 classes (500 images per class) and downsized to 64 x 64 coloured images. It consist of 500 training images, 50 validation images, and 50 test images in each class.

III. Literature Review

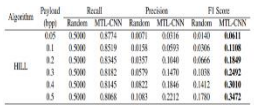
Comparative study on various subtitles

Authors &Year	Methodology or Techniques used	Advantages	Issues	Metrics used
1. **Hiroya Kato, Kyohei Osuge, Shuichiro Haruta, et al. (2020)**	Processing in advance using multiple steganographs to intentionally take a sample image on CNN-based steganalysis. Steganography was used in RCI and RSI1 times as a preliminary analysis	This work would be a first step of countermeasures against resized stego images and contributes to future studies in steganalysis and information forensics useful to deal with resized stego images created by the NNI although the state – of – art CNN based steganalysis cannot deal with them at all.	CNN acquisition performance is reduced if the image is deliberately altered by a nearby neighboring translation before steganography.	Accuracy in a model with the proposed preprocessing is up to 34.8% higher than that in the conventional model when the same steganography is embedded
2. **Zhujun Jin, Yu Yang, Yuling Chen, et al. (2020)**	It is proposed a lightweight convolutional neural network called IAS-CNN directed at the dynamic image of steganalysis. Three local steganography algorithms, such as HUGO, WOW and SUNIWARD are used to test network performance.	If the calculation power and storage space are limited, IASCNN is best suited for steganalysis	Steganalysis based on in-depth learning usually has a large number of parameters, and its suitability for adaptive steganography algorithms is weak	With a payload of 0.4, your accuracy is both higher than 80% for WOW, and both higher than 75% for SUNIWARD

3. **Zhili Xiang, Jun Sang, Qian Zhang, et al. (2020)**	Two high-level steganographic algorithms at the local domain, namely, WOW and S-UNIWARD, were used to test the effectiveness of the model.	By adding more convolutional layers in the lower part of the model, a new arrangement of convolutional layers and pooling layers was proposed, which could process the local information better than the existing CNN model s in steganalysis.	Steganalysis for content-adaptive image steganography in the spatial domain is a difficult problem	The experimental results on BOSSbase showed that the proposed CNN could obtain better steganalysis performance than YeNet across all tested algorithms when the payloads were 0.2, 0.3, and 0.4 bpp.
4. **Xintao Duan, Nao Liu, Mengxiao Gou, et al. (2020)**	A new model of Steganography Convolution Neural Network (SteganoCNN) is proposed, which solves the problem of two images embedded in a network company image and can successfully recreate two secret images. SteganoCNN has two modules, a coding network, and a recording network, while the coding network includes two coding networks.	StegaoCNN has excellent performance capabilities and can detect steganography of different types of data, such as remote-sensing images and aerial photographs.	Hiding two secret photos in one network company photo is a challenge	Test results show that the proposed steganography scheme has a maximum image load capacity of 47.92 bits per pixel, and at the same time, we can effectively avoid the detection of steganalysis tools while keeping the stego image distorted.

5. **XINTAO DUAN 1 , DAIDOU GUO 1 , NAO LIU (2020)**	A new steganography technique for high image based on in-depth reading. Discrete Cosine Transform (DCT) is used to convert confidential imagery, and the converted image is encrypted with Elliptic Curve Cryptography (ECC)	The classified image is embedded into the Host image through the SegNet deep network model, to improve steganographic capacity	The traditional steganography image method is usually securely embedded confidential information on the host image so that the load capacity is almost negligible and the quality of the steganographic image needs	The method can allocate each pixel to an image so that the relative volume of steganography is reached. Image obtained using this steganography method has high Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index

	to enhance the anti-retroviral imagery obtained.		to be improved in the Human Visual System (HVS).	(SSIM) values, up to 40dB and 0.96.
6. Qi Li, Xingyuan Wang, Xiaoyu Wang, et al. (2020)	It features a gray steganography scheme novel that is able to hide a secret image embedded in a cover image of the same size. . A secret image is encrypted using confusing encryption technology before encryption	The proposed system solved not only the problems of leaking confidential information and the color distortion of the current standard methods but also the maintenance of high embedding capacity. The level of private reconstructed images is also advantageous	The cover images used by the existing steganography system are all color images, so their light channels are damaged after embedding a large amount of confidential information.	80000 training images and 10000 test images were collected from the ImageNet database, which is used to train network models. Compared to other methods, the average prices for PSNR and SSIM for steg images can reach 0.987 and 42.3, respectively.
7. Weixuan Tang, Bin Li, Shunquan Tan. (2019)	A steganographic scheme with a novel function called adversarial embedding (ADV-EMB), which achieves the goal of hiding the stego message while at the same time deceiving the convolutional neural network (CNN) based steganalyzer. The proposed method operates under a standard distortion reduction framework	ADV-EMB adjusts the cost of image editing according to gradients and distributes from a targeted CNN steganalyzer.	The proposed steganographic system achieves better security performance compared to the target-unknowingly targeted steganalyzer by increasing the missed detection rate. If the target steganalyzer is reached by the steganographer but the steganalyst is unaware of enemy activity, a high level of miss can be achieved through adversarial embedding (ADV-EMB) to combat the target steganalyzer.	The proposed steganographic scheme achieves better security performance against the target adversary-unaware steganalyzer by increasing its missed detection rate. When the target steganalyzer is accessible by the steganographer but the steganalyst is unaware of the adversary operation, a high missed detection rate can be achieved by adversarial embedding (ADV-EMB) to counter the target steganalyzer

8. Xiangyu Yu, Huabin Tan, Hui Liang (2019)	CNN multi-tasking learning - used to train at least two related tasks simultaneously, can enhance the normal performance of each task or main task by using specific domain information contained in related job training signals. CNN's multi-tasking app is learning about the pixel-sided binary division function, estimating whether each pixel in a picture has been altered due to steganography.	It is advantageous to train the two tasks jointly, and superiors result can be attained with suitable λpixel.	It's hard to read a section where CNN hires to distinguish secret messages embedded in an image due to weak stego symbols and limited useful information.	 Algorithm / Payload (bpp) / Recall (Random, MTL-CNN) / Precision (Random, MTL-CNN) / F1 Score (Random, MTL-CNN) HILL: 0.05 — 0.5000, 0.5774, 0.0071, 0.0316, 0.0140, **0.0611** 0.1 — 0.5000, 0.5519, 0.0158, 0.0593, 0.0306, **0.1108** 0.2 — 0.5000, 0.5545, 0.0357, 0.1040, 0.0666, **0.1649** 0.3 — 0.5000, 0.5182, 0.0579, 0.1470, 0.1038, **0.2492** 0.4 — 0.5000, 0.5145, 0.0822, 0.1846, 0.1412, **0.3010** 0.5 — 0.5000, 0.4868, 0.1083, 0.2212, 0.1780, **0.3472**
9. Yafei Yuan, Wei Lu, Bingwen Feng (2017)	Focus is on the stage of image processing, 3 high-pass filtered image residuals are fed to a deep CNN architecture to make full use of the great nonlinear curve fitting capability. As known, deep learning is powerful in pattern recognition, most previous networks only use single type of filtered residuals in steganalysis, varied high-pass filtered residuals can offer stronger features for CNN.	After filtering, the remains are replaced with a residual map of multiple channels prior to training, this step may involve the combined development of CNN parameters.	The residuals are not independent at the training stage, multi-channels residuals can involve a joint optimization in the whole stage together, far more effective than one residual map	Test results prove to be an effective way to provide better acquisition performance, achieving 82.02% accuracy in the Cropped-BOSSBase-1.01 database. The accuracy of testing CNN 3-channel model decreased by 2.41% compared to training accuracy. CNN 6-channel model can achieve 78.17% accuracy, 2.56% lower than training accuracy
10. Yao Wu, Hui Li, Junkai Yi (2019)	To improve the accuracy of the acquisition, we recommend CNN unlocked from the end to remove stego image features. The end-to-end map can be	It cn minimize the negative effect of the high-pass filtering under the condition of guaranteeing the	Reduction in the time complexity will be the main problems to be solved in our future improvement.	

<table>
<tr>
<td></td>
<td>trained to read the feature feature that works best from input images to image extraction. By combining CNN's hidden layers, the extracted features can be considered as features of both embedded images and their remaining images.</td>
<td>convergence of the network</td>
<td></td>
<td>TABLE I. ACCURACY (%) OF END-TO-END CNN, XU-NET AND SRM AGAINST S-UNIWARD AND HILL
<table>
<tr><td>EMBEDDING RATE</td><td colspan="2">0.1 bpp</td><td colspan="2">0.4 bpp</td></tr>
<tr><td>EMBEDDING ALGORITHM</td><td>S-UNIWARD</td><td>WOW</td><td>S-UNIWARD</td><td>WOW</td></tr>
<tr><td>SRM</td><td>59.25</td><td>62.24</td><td>79.53</td><td>81.62</td></tr>
<tr><td>XU-NET</td><td>57.33</td><td>60.92</td><td>80.24</td><td>80.51</td></tr>
<tr><td>END-TO-END CNN</td><td>98.47</td><td>98.57</td><td>98.55</td><td>98.65</td></tr>
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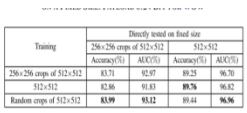
<td>11. Lionel PIBRE, Jerome PASQUET, Dino IENCO, Marc CHAUMONT
(2018)</td>
<td>Deep learning – using the same embedding key for embedding with the simulator in the different images, use a network that is in height, and only consists two convolutional layers</td>
<td>Due to the internal integration reduction and the preservation of local knowledge, the results obtained from the Convolutional Neural Network or Fully Connected Neural Network, if properly stated, exceed the normal use of EC-RM RM.</td>
<td>Problem of cover source variability, paid loading location limit issue, with the idea that the same embedding key is reused</td>
<td>Results show a more than 16% reduction in segmentation error with our CNN or FNN.</td>
</tr>
<tr>
<td>12. Jaeyoung Kim, Hanhoon Park,Jong-Il Park
(2019)</td>
<td>It uses CNN and is actually based on the assumption that the pixel variation caused by embedding images on cover images using a steganographic image method will be statistically different from that caused by embedding stego images using the same method.</td>
<td>With the proposed dual channel or CNN dual network, we can use additional input images generated by embedding additional data into the image input, which enabled CNN to read.</td>
<td>The method was dealing with color images only and not gray scale images. The reliability of their results was low. Method was verified in 1-LSB only</td>
<td>The proportional divisions of the proposed CNN dual network have increased significantly and were 4-6% higher than those of regular CNN.</td>
</tr>
<tr>
<td>13. Ying Zou, Ge Zhang, Leian Liu
(2019)</td>
<td>This paper proposes to transfer helpful information from a high-level embedded image to a CNN model training to obtain a low-level embedded image. The</td>
<td>The proposed model is very effective in capturing mathematical features such as steganalysis-</td>
<td>Additional statistical imagery of the world is required to make the image changes caused by steganography are very</td>
<td>Test results on the BOSSbase website show that migration-based learning significantly improves CNN-based</td>
</tr>
</table>

	method of conveying information is the same as previous training without the use of additional helpful features.	related neighborhoods, and automatically learns the effective presentation of steganalysis features, greatly reducing the need for personal knowledge and time.	important in the space feature of the image.	steganalysis detection capabilities at low embedding rates.
14. Hyeokjoon Kweon, Jinsun Park, Sanghyun Woo, et al. (2021)	We present the secret key concept of the steganography problem for many images. The model can provide direct access to private image only to authorized people. The model makes it possible to extract a stolen stolen image without exposing other hidden images	The secret key concept is expanded in multi-image steganography, which hides multiple secret images within a single cover image	Typical methods tend to leak confidential information because they do not provide access to the private image and often hide all confidential information at once.	Images are resized to 256 × 256 using bicubic interpolation. The proposed model is trained 100 times with Adam optimizer with a reading rate of 1 × 10−3. The learning rate is multiplied by 0.2 when the loss ceases to decrease.
15. Weike You, Hong Zhang, Xianfeng Zhao (2020)	To test the possibility of network exploitation by combining images of various sizes without retraining its parameters. On the assumption that the sound of the natural image is the same between the areas under the images using in-depth reading	CNN based steganalysis methods eliminate the need for hand-crafted features and automatically extract more extensive features from data through the use of backpropagation.	Software programs attempt to make the payload look like random image noise produced by the camera sensor and circuitry; detection of imagery associated with criminal activities using the human eye alone is impossible.	
16. Chaoning Zhang, Chenguo Lin, Philipp Benz, Kejiang Chen, Weiming	The formulation of deep concealment is presented first, followed by a discussion of the several basic meta-architectures. A comprehensive review of its three uses, namely safe steganography, light field	In essence, this strategy is similar to noise augmentation, but it uses a plus and minus operation to circumvent the non-differentiability issue. This method achieves	Grid search for selecting the embedding scheme is time consuming. Message is used rather than images.	SSIM and MSE 0.9794 for stego images and 0.9842 for secret images.

Training	Directly tested on fixed size			
	256×256 crops of 512×512		512×512	
	Accuracy(%)	AUC(%)	Accuracy(%)	AUC(%)
256×256 crops of 512×512	83.71	92.97	89.25	96.70
512×512	82.86	91.83	**89.76**	96.82
Random crops of 512×512	**83.99**	**93.12**	89.44	**96.96**

Zhang and In So Kweon 2021	messaging, and robust watermarking, was undertaken with the goal of hiding a hidden message in the image carrier.	SOTA performance for JPEG attack robustness and has also been demonstrated to perform well with video compression formats like MPEG and XVID.		
17. Nandhini Subramanian 2020	The goal of this project is to create a steganography model for embedding the secret image and a steganalysis tool for extracting the secret image. A novel auto encoder-decoder with a deep convolutional neural network is proposed in this paper. A subset of the COCO, CelebA, and ImageNet datasets is used for training and testing.	Higher invisibility, security, and robustness have been demonstrated using the proposed strategy.	Though the issue of hiding capacity, which is a problem with most old methods, has been resolved. Because of the complexity of the models, the capacity is raised at the expense of storage space, memory, and calculation time.	For the COCO, CelebA, and ImageNet datasets, the PSNR value of the embedding network on testing data that is not exposed to the model during training is 31.96, 32.26, and 31.18, respectively. Similarly, the extraction module's PSNR value for COCO, CelebA, and ImageNet datasets is around 27.9.
18. Shao-Ping Lu, Rong Wang, Tao Zhong and Paul L. Rosin 2019	This paper presents an Invertible Steganography Network for picture steganography in this research, in which the same network's forward and backward propagation operations are used to embed and retrieve concealed images, respectively.	We can efficiently generate both the container image and the exposed hidden image(s) with excellent quality by sharing all characteristics of our single ISN architecture. Furthermore, by organically expanding the number of channels in the hidden image branch, the capacity of picture	Both recommended strategies do not work well with StegExpose detection. It's also worth noting that the detection curve for a previously established method is marginally better than ours, although the PSNRs measurements given earlier are the inverse.	The photos are objectively evaluated using the PSNR (Peak Signal to Noise Ratio) and SSIM (Structural Similarity) metrics. The PSNR and SSIM measurements for the container image gain +12.06 dBs and +0.115, respectively.

		steganography is greatly improved in our architecture.		
19. Hugo Ruiz, Marc Chaumont, Mehdi Yedroudj, Ahmed Oulad Amara , Fr´ed´eric Comby1, and G´erard Subsol 2020	In this study, we confirm that error law also plays a role in steganalysis, and this is in the context of a borderline, i.e., medium-sized network on a large, limited, and diverse site, after an in-depth discussion of targeted behavior. CNN as a function of their size and website size.	Low Complexity Network CNN has the advantage of having a significantly reduced number of parameters (twenty times fewer parameters than SRNet). Because the number of parameters is reduced, it takes less time to learn and converges faster to an optimal solution.	When scaling the database, the network should be far enough away from the random-guess zone to see a tangible improvement in performance. As a result, we must select a payload such that the LC-Net accuracy is well below 50%. This is difficult because there are no findings for LC-Net on "controlled" databases with a lot of variation, such as LSSD.	The accuracy of the payload load of 0.2 bpnzacs indicates that the larger reading set (100k, 200k, 1M), the better. The accuracy of the 20k website is 62 percent, and it improves by about 2% each time the reading set size is doubled.
20. Olivia Byrnes, Wendy La, Hu Wang, Congbo Ma, Minhui Xue and Qu Wu 2019	This survey summarizes current advances in data hiding strategies for watermarking and steganography, categorizing them based on model architectures and noise injection methods. The objective functions, assessment measures, and datasets that were used to train these data hiding models are all outlined in detail.	This technique has the advantage over previous standard algorithms in that it does not require expert programming expertise and can be retrained for different applications and attack types rather than having to be built from scratch.	Attention techniques to increase embedding could all be combined in a future model to provide a very resilient framework. However, in practice, the added overhead of having these extra features could be a concern.	U-Net (CNN) can generate high-quality pictures for messages up to 100 bits with a 95% recovery accuracy rate.
21. FuZhangjie, LiEnlu ,ChengXu, HuangYongfeng,and HuYuting 2021	This research examines three features of deep learning-based image steganography systems in depth. The first is to present two approaches to cover picture acquisition: generative adversarial	This method can do image steganography without using a carrier image, which has obvious advantages in terms	Based on deep learning, adversarial samples have proven resilience to steganalyzers. The generalization of adversarial examples, however, has to be improved in the face of	Despite the fact that steganalysis accuracy is decreasing as the network improves, the secret carrier's resistance to steganalysis is still

	networks and adversarial samples. The second goal demonstrates how to build steganographic distortions. The final step is to explain how to generate stego images using the encoding decoding network.	of avoiding statistical analysis.	multiple types of steganalyzers and steganalyzers trained on adversarial examples.	relatively weak, with an accuracy of 71%.
22. Ching-Chun Chang, Xu Wang, Sisheng Chen, Isao Echizen, Victor Sanchez, and Chang-Tsun Li **2019**	In particular, neural networks are used in an analytics module to learn about data distribution, with a pre-existing mechanism handling the remaining responsibilities. In this research, we look at the modular framework and show how to use deep neural networks in a reversible steganographic system called prediction-error modulation, in which an analytics module is used to predict pixel intensity.	This modular framework, which fully utilises the traditional coding module while deploying powerful neural networks in the analytics module, should be more dependable than an end-to-end framework.	Input images are just concatenated. Shared generator and discriminator. User has to choose seed value instead of cover image.	When compared to the ground-truth with a PCNR score of roughly 40, RCAN and RDN are more able to recall textural areas with reasonable precision.
23. Arivazhagan Selvaraj, Amrutha Ezhilarasan , Sylvia Lilly Jebarani Wellington, Ananthi Roy Sam **2020**	Breakthroughs in the field of digital picture steganalysis are highlighted in this study of convolutional neural network-based deep learning architectures. To anticipate spatial domain steganographic algorithms, multiple CNNs are first created. Through diligent designs integrating adjustments to activation functions, pooling, pre-processing filters, and residual connections, such	For transform domain steganalysis, CNN architectures have also been built and have outperformed classical models. With the introduction of GPUs, the computational complexity of current CNN architectures has been considerably lowered.	Despite the fact that CNN models have made significant progress in recognising cover or stego images of specific or multiple stego algorithms, the active steganalysis process of location and payload extraction from stego photos with CNN models is still in its early phases.	In comparison to the accuracy of 54.7 percent achieved by XuNet models, the proposed model achieves a 74.5 percent accuracy.

	structures outperformed classic steganalysis approaches.			
24. Ching-Chun Chang 2021	The prior estimation using a standard linear predictor is modified using a neural network model in this paper. The refinement could be considered a low-level vision activity to some extent (e.g., noise reduction). The biological validity of a cutting-edge neuroscience-inspired low-level vision model based on long short-term memory is investigated in this method.	The visual quality of posterior photographs is noticeably greater than that of prior images, particularly at the borders and in textural areas.	The height of the peak bin on the posterior histogram is not necessarily higher than the height of the same bin on the prior histogram, which is quite striking.	PSNR values of around 50 with all errors falling under 95 percentile category.
25. Hyeokjoon Kweon, Jinsun Park, Sanghyun Woo and Donghyeon Cho 2020	The concept of secret key is extended to the steganography of many images in this study, which hides many secret images within a single cover image. The steganography model starts with a bunch of secret photos and cover images, and then produces a picture of a container with secret keys in each secret photo.	Unlike earlier picture steganography systems, this model allows only authorised users to see the secret image. Furthermore, this model allows only the quaried secret picture to be extracted without revealing other hidden images.	Computation is increased by using additional images.	The suggested model is trained with an Adam optimizer with a learning rate of (1 *10^(-3)) over 100 epochs. When the loss has ceased falling, the learning rate is multiplied by 0.2.
26. Bisma Bashira and Arvind Selwalb 2021	This research examines and compares a number of cutting-edge deep learning-based image steganalysis algorithms. Furthermore, the different open research issues that may provide a future direction in this lively subject of picture	By combining it with BOWS2 and integrating data augmentation practise on the BOSSbase, the dataset is further increased. In comparison to existing conventional	To solve the tradeoff between training time and model architectural complexity, domain adaptability and light-weight models must be investigated.	The steganographic algorithm S-UNIWARD has an accuracy of 57.33% on 0.1 bpp and 80.24% on 0.4 bpp for the suggested modelOn the payload of 0.1 bpp, HILL's accuracy is

	steganalysis have been identified. The lack of large-scale data for training CNN models is a major barrier for deep learning-based techniques, according to this investigation.	models, the network's performance with selection channel awareness is substantially superior.		58.44%, while on 0.4 bpp, it is 79.24%
27. Shamal Salunkhe and Surendra Bhosale 2020	CNN models for steganalysis are developed in this research. CNN are used to extract spatio-temporal information or characteristics, as well as to classify them. With the same dataset, the results of steganalysis with AlexNet and SRNeT have been compared. The stegnalytic error rates for various payloads are compared.	Image is secret message. Basic and minimum architecture is used. Domain knowledge is not required.	In detection, nonlinear residuals created during embedding are rarely used. One of the limitations of steganalysis is the diminishing of characteristics. In JPEG steganalysis, variable embedding rate can be investigated.	Using WOW algorithm at a payload of 0.4 the detection error rate of Alexnet is 0.178 and SrNet's is 0.086. Using S-UNIWARD algorithm at a payload of 0.4 the detection error rate of Alexnet is 0.269 and SrNet's is 0.125.
28. Abhishek Das, Japsimar Singh Wahi, Mansi Anand and Yugant Rana 2020	This research seeks to use deep neural networks to encode and decode several hidden images under a single cover image of the same resolution, as well as to execute multi-image steganography, which involves hiding three or more images in a single cover image.	Better performance and security than the previously proposed LSB methods. It is highly secure and robust with increased hiding capacity.	They relied significantly on visual perception for total loss and did not test various sorts of losses that would have been more appropriate for their model. A new way for validating the approach's strength would aid in improving the results in the proper direction.	Despite the fact that steganalysis accuracy is decreasing as the network improves, the secret carrier's resistance to steganalysis is still relatively weak, with an accuracy of 65%.
29. Jiabao Cui, Pengyi Zhang, Songyuan Li, Liangli Zheng,	Multitask Identity-Aware Image Steganography (MIAIS) is proposed in this paper to gain direct recognition for container images without restoring	The trials reveal that MIAIS is more effective than other visual information hiding methods, that the recognition results	More mosaics and lattices are introduced in the ways that do not lose content. However, the loss of content will result in some artefacts, lowering the visual quality.	On the AFD dataset, recognition accuracy varies between 0.05 % and 0.28 %, and 0.04 percent and 0.30 % on the TinyImageNet

Cuizhu Bao, Jupeng Xia, and Xi Li 2019	confidential images. The big problem with direct recognition is to keep the identity details of the private photos in the container images while making the container images look like cover photos. In order to deal with conflicting elements, we introduce simple content losses to store identity information and design minimal configurations.	are more consistent across different cover datasets, and that MIAIS outperforms state-of-the-art high-capacity picture steganography approaches on face identification.		dataset. On average, the MSE, MS-SSIM, and PSNR all change within 0.06 %, 0.16 %, and 0.60 %, indicating that the results may be transferred across datasets.
30. Rini Elizabeth Cherian 2021	The paper proposes an image "Steganography" approach that is far more secure. Despite the fact that it employs some standard "Steganography" processes, combining them with "cryptography" and "neural" organisations makes it difficult to crack. The encryption layer adds an extra layer of security.	Generation of more realistic images. Highly Secure	No explanation on how to uncover the secret message. If generative model fails then the secret image is lost.	For stowing away and uncovering design portion, Algorithm uses CNN in support of Adam analyzer, resulting in exactness of around 90% at train and test datasets.

IV. PROPOSED ALGORITHM

The system that we have implemented in this project uses a CNN to hide an image inside another image hence making the secrete image effectively invisible to the observer. The neural networks that we have used in our work determines where in the cover image it can hide the information and hence, is a more efficient process than LSB manipulation. We have also trained a decoder network to reveal the secret image from the container image. This process is done in such a way that the container image does not change much and the changes to the container image are not discernible.

It can be safely assumed that the intruder does not have access to the original image. The secret image is effectively hidden in all 3 colour channels of the container image.

3 different activation functions were used to test the network along with various learning rates.

The activation functions used were:

1. Rectified Linear Unit (RELU) {de-facto for most CNN networks}

2. Tanh

3. Scaled Exponential Linear Unit (SELU) {has been used to avoid the vanishing gradient problem caused by RELU activation function}

Advantages: -

1. CNNs learn the most efficient way to hide the secret image inside the container image which is unpredictable for us. As a result, it adds to the security of steganography by making the process unpredictable.

2. The container image is note altered a lot which makes the changes harder to spot.

3. The changes can only be spotted by the decoder network that is specifically trained to do it.

4. The process is flexible that means that before feeding to network, some changes can be made to the image to make the process more secure.

Disadvantages: -

1. This is not a full proof method and the changes can be spotted by a specially trained network and the original image can be somewhat retrieved.

We have chosen this method because the advantages clearly outweigh the disadvantages

V. EXPERIMENTS RESULTS

ImageNet is a massive dataset that contains images from the WordNet hierarchy, with each node holding hundreds of images. The image on ImageNet has no copyrights and just contains links or thumbnails to the original image. Images of various sizes make up the dataset. The number of photographs, the classes they belong to, the background, and the image size can all be customized based on the requirements.

ImageNet's Tiny ImageNet dataset is a subset of the larger ImageNet dataset. The dataset consists of 100,000 photos divided into 200 classes (500 images per class) that have been reduced to 6464 coloured images. There are 500 training images, 50 validation images, and 50 test photographs in each class.

The quality of image steganography is assessed using a variety of approaches. Each of these methods evaluates a different component of the steganographic outcome. Mean Square Error (MSE), Peak Signal to Noise Ratio (PSNR), and Structured Similarity Index Measure (SSIM) are some of the well-known approaches.

Mean Square Error (MSE):-

The averaged value of the square of the pixel-by-pixel difference between the original image and the stego-image is the Mean Square Error. It provides a measure of the cover image error caused by the data embedding procedure. A lower MSE value denotes a high-quality embedding.

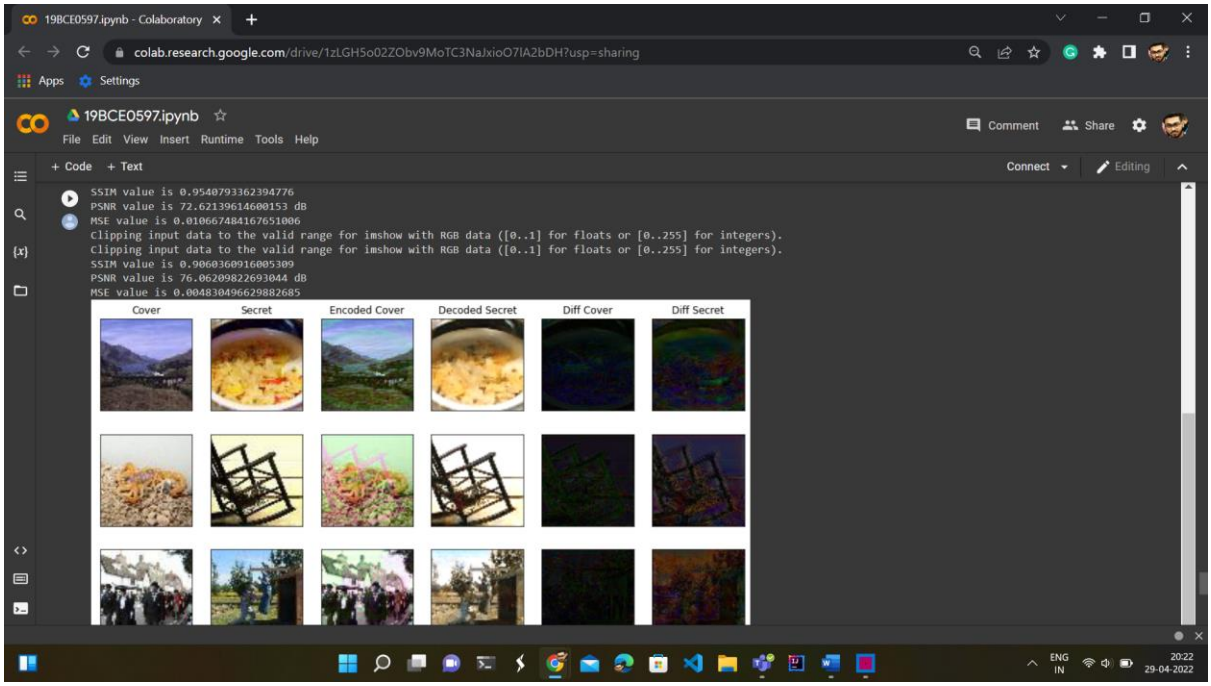
Peak Signal to Noise Ratio (PSNR):-

PSNR is another useful metric for determining the degree of embedding-induced distortion in the cover image. It's the ratio of a signal's greatest possible value to the power of distortion noise (MSE). It is expressed in decibels (dB). A higher PSNR value denotes a higher-quality embedding.

Structured Similarity Index Measure (SSIM):-

SSIM is a comparison statistic used to see how similar the cover picture and the stego-image are. The perceived difference between the two images is measured.

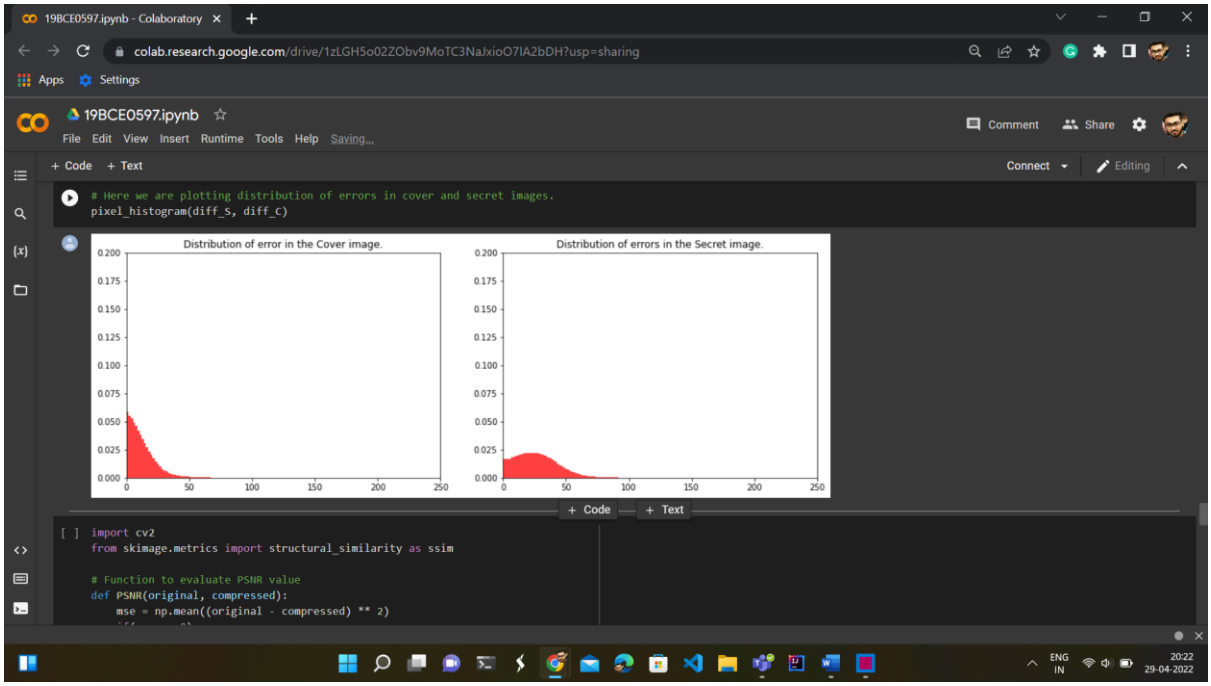
We were able to achieve MSE value of 0.006259255611669444, PSNR value of 76.21412244818211 dB, and SSIM value of 0.9651267817491244.



Pixel wise average errors for Secret Image were 27.152718 on a scale of 256.

Pixel wise average errors for Cover Image were 20.411884 on a scale of 256.

Along with above distribution of errors in cover and secret images has been shown below.



The loss through epochs has been shown below:-

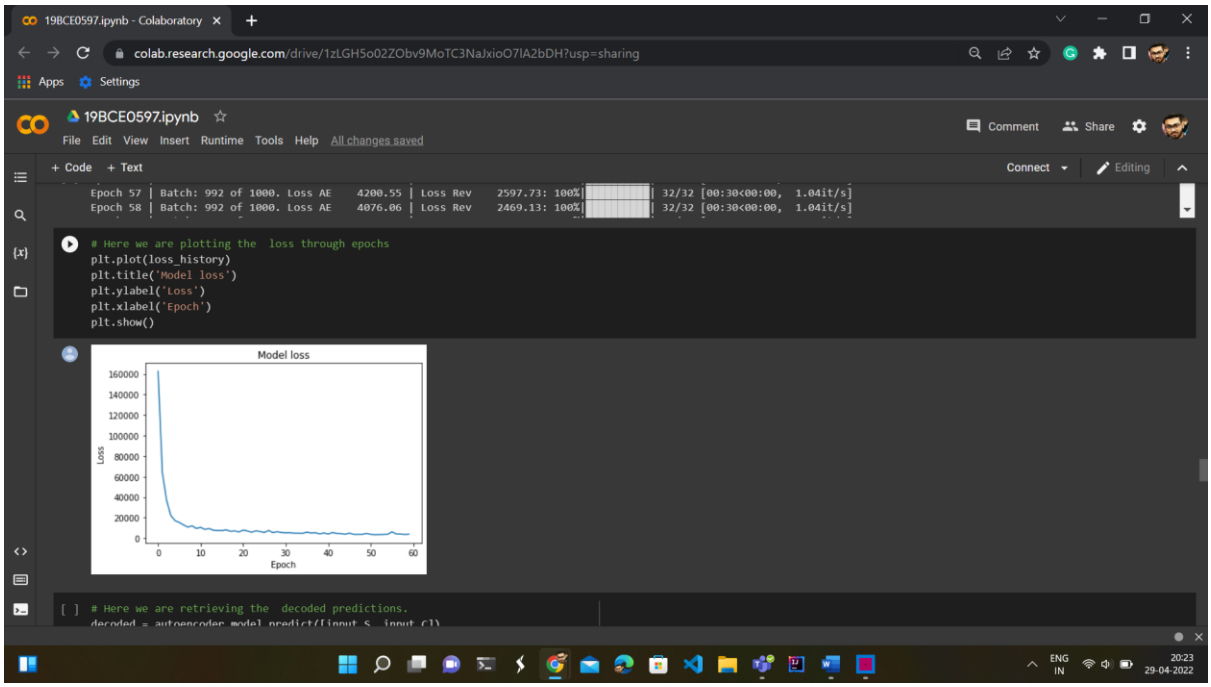
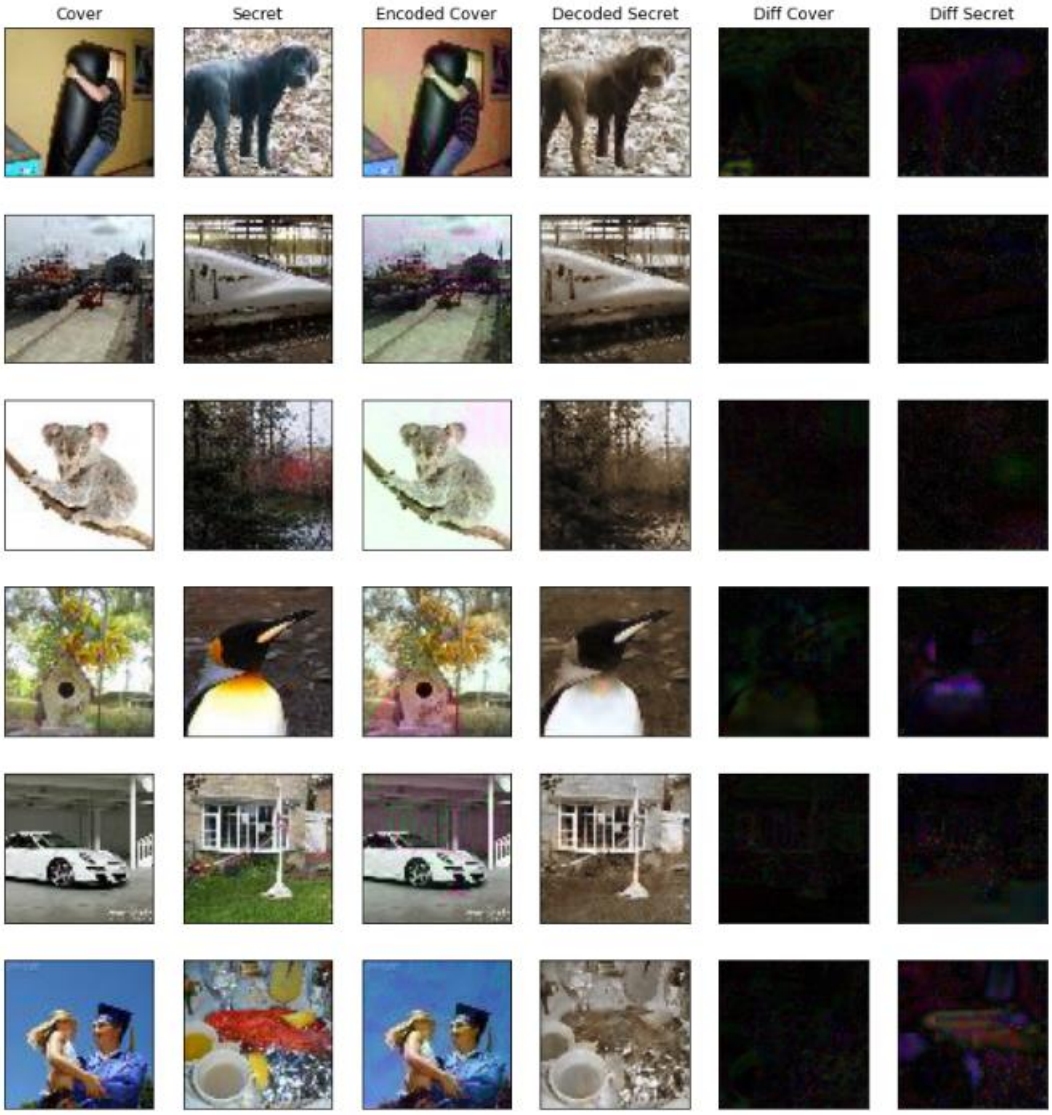


Image below shows the cover image, secret image, container image, revealed image respectively. RELU activation function produced the best results and SELU activation function produced the worst results with different activation functions. Given below are the results obtained on using different activation functions.

1. RELU


Cover
Secret
Encoded Cover
Decoded Secret
Diff Cover
Diff Secret

SSIM value is 0.8939719109993859

PSNR value is 72.71192127985658 dB

MSE value is 0.010447430573533138

SSIM value is 0.9244448899031624

PSNR value is 76.21412244818211 dB

MSE value is 0.004664331104070731

SSIM value is 0.92911087801986

PSNR value is 74.67211487481308 dB

MSE value is 0.006652580734333079

SSIM value is 0.9651267817491244

PSNR value is 73.82109884139598 dB

MSE value is 0.008092668651015305

SSIM value is 0.9280019574836267

PSNR value is 74.93678898322835 dB

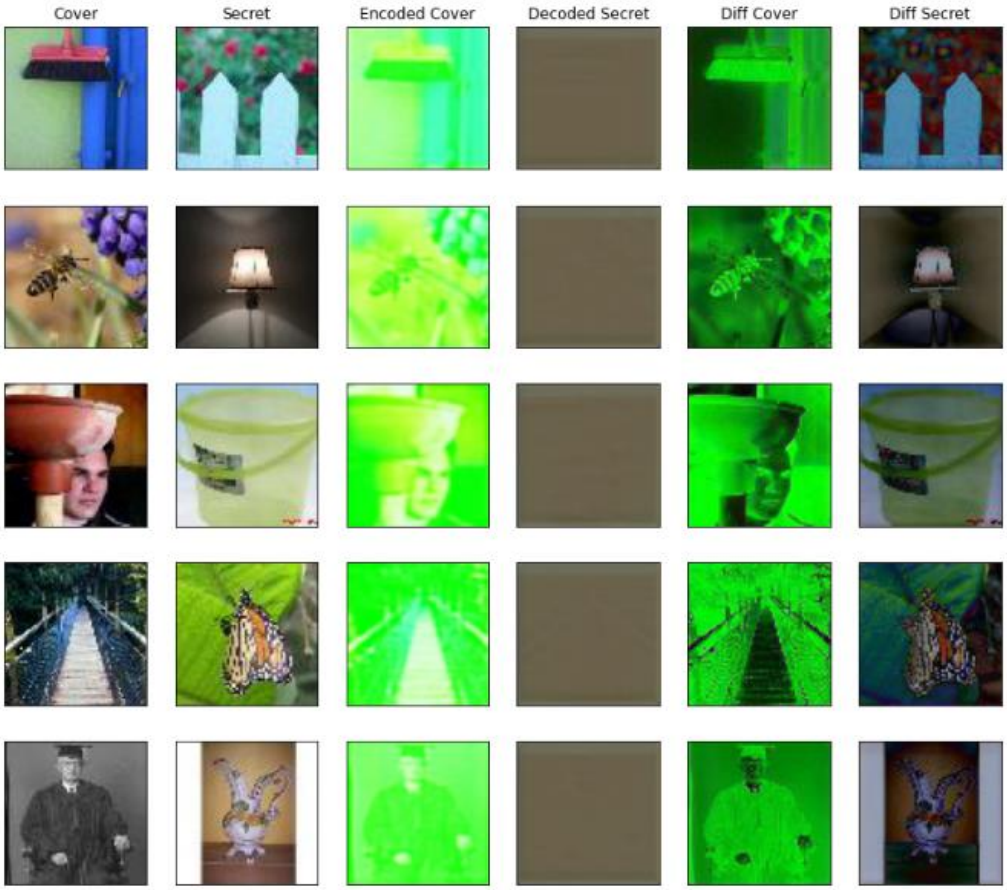
MSE value is 0.006259255611669444

SSIM value is 0.897791248823328

PSNR value is 74.78361509242083 dB

MSE value is 0.006483956730341666

2. TANH


Cover
Secret
Encoded Cover
Decoded Secret
Diff Cover
Diff Secret

SSIM value is 0.7249706379747698

PSNR value is 57.778771985527754 dB

MSE value is 0.32533022816724777

SSIM value is 0.6853665491588302

PSNR value is 57.97070994579716 dB

MSE value is 0.3112652170478809

SSIM value is 0.6125868370464277

PSNR value is 55.291631894645086 dB

MSE value is 0.5768175525002196

SSIM value is 0.5114901347567362

PSNR value is 55.76227153024707 dB

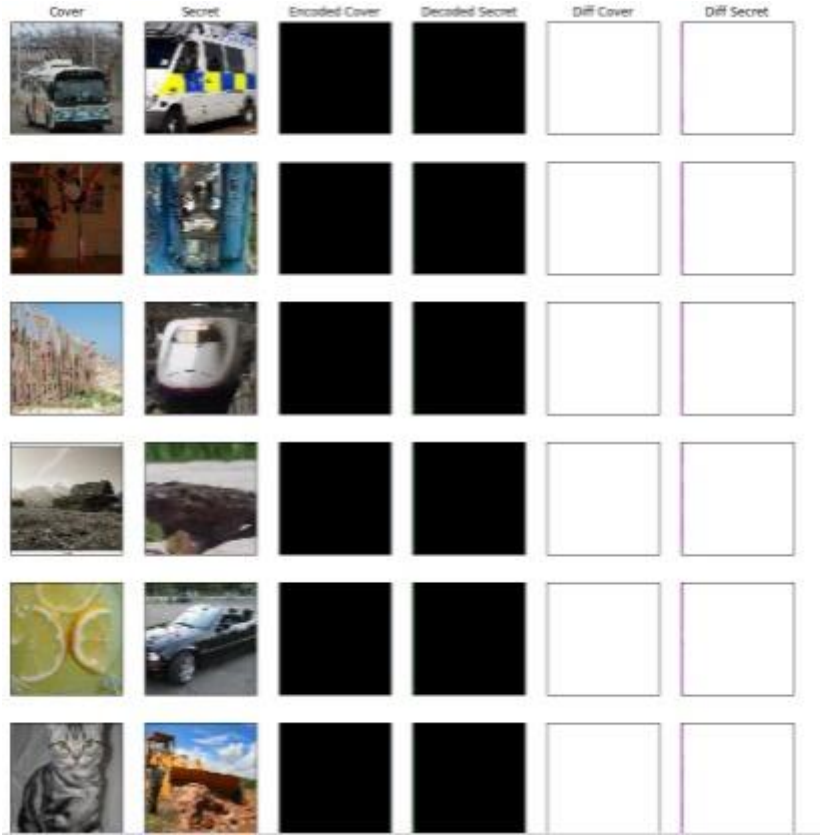
MSE value is 0.5175764130335532

SSIM value is 0.718185592713867

PSNR value is 56.32913410752248 dB

MSE value is 0.45424294067568477

3. SELU


Cover
Secret
Encoded Cover
Decoded Secret
Diff Cover
Diff Secret

SSIM value is -0.0749665991475593

PSNR value is 41.390807342395384 dB

MSE value is 14.161878601022435

SSIM value is -0.09671444293532629

PSNR value is 42.51343772482506 dB

MSE value is 10.935983550429427

SSIM value is -0.21655353029185795

PSNR value is 40.57839780171014 dB

MSE value is 17.07504446106052

SSIM value is -0.22549568482169927

PSNR value is 40.96982659191187 dB

MSE value is 15.603390647455749

SSIM value is -0.3066082760567415

PSNR value is 40.92396672988879 dB

MSE value is 15.7690285743913

SSIM value is -0.1350593767657198

PSNR value is 41.3890867141432 dB

MSE value is 14.16749116299357

VI. COMPARATIVE STUDY

In this section the results of the proposed methodology are compared with the existing.

Parameter	**[Paper 1]**	**[Paper 2]**	**[Paper 3]**	**Proposed Algorithm**
Dataset	Lena and Baboon	ImageNet	1 RGB image	Tiny ImageNet
Method Used	GAN	GAN	LSB	CNN
Architecture	Traditional Method	U NET	Traditional Method	Encoding and decoding
MSE	0.13	-	-	0.006
PSNR	56.95	40.66	62.53	76.214
SSIM	-	0.964	-	0.965

Paper 1: A. Arya and S. Soni, ‘‘Performance evaluation of secrete image steganography techniques using least significant bit (LSB) method,’’ Int. J. Comput. Sci. Trends Technol., vol. 6, no. 2, pp. 160–165, 2018.

Paper 2: X. Duan, K. Jia, B. Li, D. Guo, E. Zhang, and C. Qin, ‘‘Reversible image steganography scheme based on a U-Net structure,’’ IEEE Access, vol. 7, pp. 9314–9323, 2019.

Paper 3: K. A. Al-Afandy, O. S. Faragallah, A. Elmhalawy, E.-S.-M. El-Rabaie, and G. M. ElBanby, ‘‘High security data hiding using image cropping and LSB least significant bit steganography,’’ in Proc. 4th IEEE Int. Colloq. Inf. Sci. Technol. (CiSt), Oct. 2016, pp. 400–404.

PSNR Value

Method	Dataset	PSNR Value
High security data hiding using image cropping and LSB least significant bit steganography	RGB Image	62.53
Performance evaluation of secrete image steganography techniques using least significant bit (LSB) method	Lena and Baboon	56.95
Reversible image steganography scheme based on a U-Net structure	ImageNet	40.66
Generative steganography with Kerckhoffs' principle based on generative adversarial networks	USC-SIPI	64.7
LSB based image steganography using dynamic key cryptography	Lena	32.09
Our implemented method	Tiny ImageNet	76.214

SSIM Value

Method	Dataset	SSIM Value
Provably secure generative steganography based on autoregressive mode	ImageNet	0.64
Reversible image steganography scheme based on a U-Net structure	ImageNet	0.964
Our implemented method	Tiny ImageNet	0.965

MSE Value

Method	Dataset	MSE Value
Performance evaluation of secrete image steganography techniques using least significant bit (LSB) method	Lena and Baboon	0.13

Coverless steganography for digital images based on a generative mode	DTD and COCO2017	0.354
Our implemented method	Tiny ImageNet	0.006

VII. CONCLUSION AND FUTURE WORK

- *The network was constructed and is performing well. Even though it has shown good performance encoding and decoding, it is not a full proof system as every technology has its shortcomings.*

- *As stated in the disadvantages, there is a lot of scope of improvement in this project as there is a chance that networks can be trained specifically for detecting that an image has been hidden inside a given image.*

- *This project can be improved upon by creating a better architecture or by altering the architecture of this network and improving its performance further, making the secret image tougher to decode by any program other than the decoder and to make it more difficult to detect the presence of any secret image inside the encoded image.*

CODE LINK:

https://github.com/ritweekraj2802/IMAGE-STEGANOGRAPHY-USING-CNN.git

VIII. REFERENCES

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